



BTCSE022401: COMPUTER ORGANIZATION AND ARCHITECTURE							
Department of Computer Science and Engineering							
Lect.	Tut.	Pract.	Credits	Evaluation Scheme			
				Component	Exam	Weightage	Pass
3	-	-	3	100 Marks Theory/ Practical	FET	20	40
					MSE	30	
					ESE	50	40
Prerequisites: • Basic knowledge of computer fundamentals and computer system components • Understanding of number systems (binary, decimal, hexadecimal) • Familiarity with digital logic concepts such as logic gates, combinational and sequential circuits • Basic understanding of data representation in computers • Introductory knowledge of programming concepts (preferably C) • Awareness of operating system basics, including processes and memory concepts							
Course Outcomes: After the end of this course students will be able to							
CO1	Describe the basic concepts of computer organization and architecture, including functional units, computer components, and CPU structure.						
CO2	Apply instruction set concepts such as instruction formats, addressing modes, assembly language, instruction execution, and distinguish between RISC and CISC architectures.						
CO3	Perform integer and floating-point number representation and arithmetic operations used in the arithmetic and logic unit.						
CO4	Analyze internal and external memory organization, including cache memory, virtual memory, and secondary storage devices.						
CO5	Evaluate control unit operations, input/output organization, instruction pipelining, and multiprocessor systems with cache coherence mechanisms.						

Course Articulation Matrix: Mapping of Course Outcomes (COs) with Program Outcomes (POs)

Course Outcomes (COs) / Program Outcomes (POs)	1	2	3	4	5	6	7	8	9	10	11	12	PSO 1	PSO 2
CO1														
CO2														
CO3														
CO4														
CO5														

3- High Contribution, 2- Medium Contribution, 1-Low Contribution

Syllabus (Theory)		
Units	Description	Hours
I.	Introduction: Concept of computer organization and architecture, Fundamental unit, Computer function and interconnection, CPU structure and function.	5
II.	Instruction Sets: Characteristics, Types of operands, Types of operations, Assembly language, Addressing modes, Instruction format, Types of instruction, Instruction execution, Machine state and processor status, Structure of program, Introduction to RISC and CISC architecture.	5
III.	Computer Arithmetic: The arithmetic and logic Unit, Integer representation, Integer arithmetic, Floating point representation, Floating point arithmetic, Introduction of arithmetic co-processor.	5
IV.	Memory Organization: Internal Memory: Semiconductor main memory, Error correction, Advanced DRAM organization, Virtual memory systems and cache memory systems. External Memory: Organization and characteristics of magnetic disk, Magnetic tape, Optical memory, RAID, Memory controllers.	7
V.	Control Unit: Control unit operation: Micro-operations, Control of the processor, Hardwired implementation, Micro-programmed Control Unit, Basic concepts, Micro-instruction sequencing, Micro-instruction execution, Applications of micro-programming.	7
VI.	Input/ Output Organization: External devices, I/O module, Programmed I/O, Interrupt driven I/ O, Direct memory access, I/O channels and processors, External interface. Instruction pipe-lining: Concepts. Parallel processing: Multiple processor organization, Symmetric multiprocessor, Cache coherence and the MESI protocol.	7
Textbooks:		
1. William Stalling, Computer Organization and Architecture: Designing for Performance, Prentice Hall Publication, 8th Edition, 2009. 2. Hayes, Computer Architecture and Organization, McGraw-Hill Publication, 3rd Edition, 2012. 3. Zaky, Computer Organization, McGraw-Hill Publication, 5th Edition, 2011		

References :

1. Hennessy and Patterson, Computer Architecture: A Quantitative Approach, Morgan and Kaufman Publication, 4th Edition, 2007.
2. Morris Mano, Computer System Architecture, Pearson Education India, 3rd Edition, 2007.
3. Mostafa Abd-El-Barr, Hesham El-Rewini, Fundamentals of Computer Organization and Architecture, Wiley Publication, 1st Edition, 2004.
4. Miles J. Murdocca, Vincent P. Heuring, Computer Architecture and Organization: An Integrated Approach, Wiley Publication, 1st Edition, 2007.
5. Sajjan G. Shiva, Computer Organization, Design, and Architecture, CRC Press, 5th Edition, 2013.

E- Reference Links:

E- Reference Links (COA): YouTube Links:

1. https://www.youtube.com/watch?v=OUNjl_1_5GY&list=PL3rE2jS8zxAwnA5uujTHmwLhK7dqEgYy
2. <https://www.youtube.com/watch?v=7UmYfl9SqfQ>
3. <https://www.youtube.com/watch?v=ZW1gb3h-f9k>
4. <https://www.youtube.com/watch?v=zwovvWfkuSg>
Types of Operands, Numbers, Characters, Logical Data.
1. <https://www.youtube.com/watch?v=QSdGRSvyf6k>
2. <https://www.learncomputerscienceonline.com/instruction-format/>

E- Reference Links (MCQ COA Quiz):

1. [Top 55 Computer Organization and Architecture MCQs](#)
2. [Computer Architecture MCQ \(Multiple Choice Questions\) - Sanfoundry](#)
3. [Computer Architecture MCQ Questions & Answers | Computer Science](#)